TUMOR LOCALIZATION · CHIP-DESIGN STUDY

Frequency Reduction

How narrow can the measurement band go before localization degrades?
CNN classification · raw S-parameters · LOSO · measured A3 phantom data

1–5 GHz

recommended band

1/2

the bandwidth of the full sweep

100%

F5 accuracy retained (full array)

Peter Martin · University of Utah · Electrical & Computer Engineering

Smaller bandwidth, simpler chip

An integrated version of this system lives or dies on front-end complexity: a narrower sweep means simpler synthesizers, antennas and calibration. The question is what accuracy that costs, and where the band should sit.

1. **WHERE does the information live?** slide a 2 GHz window across 0.1–8 GHz; full 4-antenna array
2. **HOW WIDE must the band be?** expand 2 → 3 → 4 → 5 GHz around the winning region
3. **Does it combine with reduced hardware?** winning band × fewer antennas / reflection-only
4. **Is the band choice robust?** repeat the whole sweep with a single antenna (S11 only)

Every cell: CNN classifier, raw mag+phase input, 100-epoch leave-one-session-out, per-position vote.

STEP 1 · BAND PLACEMENT

A 2 GHz window: the low-mid band wins

Band (GHz)	Width	Empty	A3 + F4	A3 + F5
0.1 – 2	2 GHz	99.4	96.2	90.9
2 – 4	2 GHz	**99.4**	**97.4**	**92.9**
4 – 6	2 GHz	99.4	89.1	71.7
6 – 8	2 GHz	99.4	97.4	69.7
0.1 – 8 (full)	7.9 GHz	99.3	97.4	100.0

All 16 S-parameters. LOSO per-position vote accuracy (%). Highlighted row = best 2 GHz window (mean across phantoms).

The 2-4 GHz window wins. Above 4 GHz, F5 collapses to ~70%: the large lossy glandular insert attenuates high-frequency energy before it reaches interior positions. The empty phantom is band-agnostic.

STEP 2 · BANDWIDTH

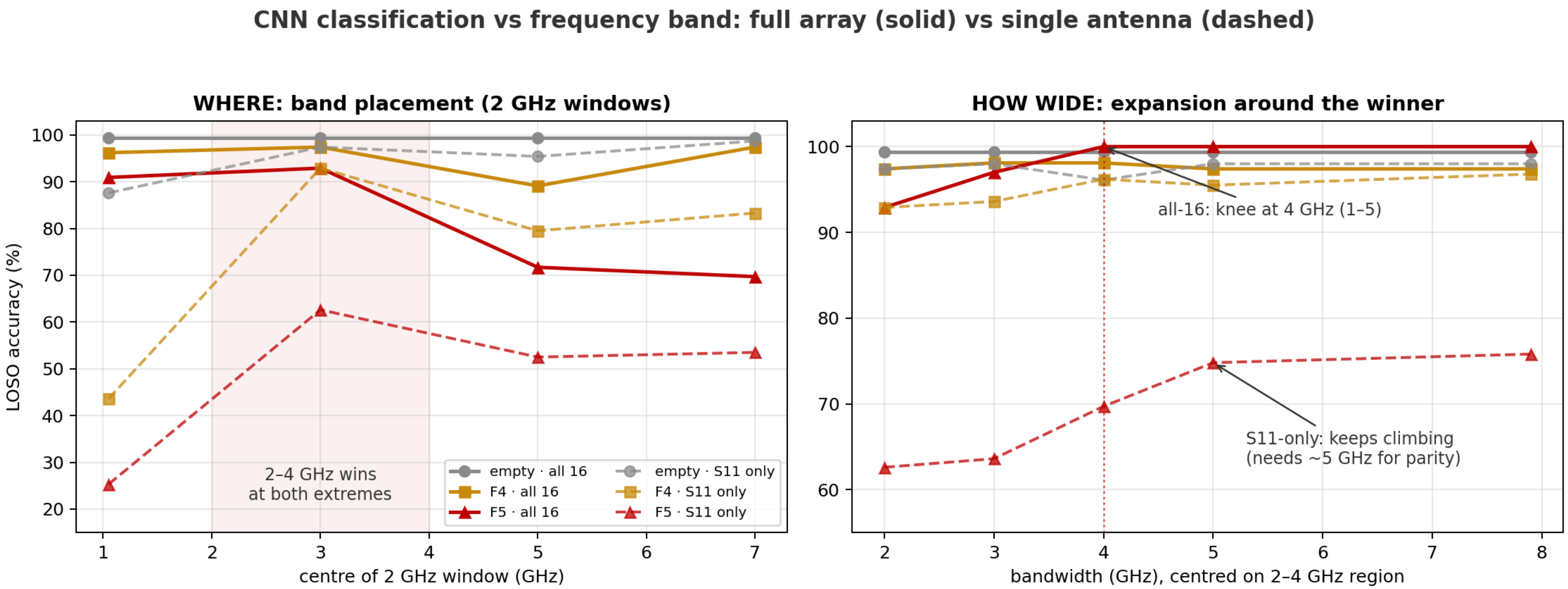
Expanding around 2-4 GHz: the knee is at 4 GHz

Band (GHz)	Width	Empty	A3 + F4	A3 + F5
2 – 4	2 GHz	99.4	97.4	92.9
1.5 – 4.5	3 GHz	99.4	98.1	97.0
1 – 5	4 GHz	**99.4**	**98.1**	**100.0**
0.5 – 5.5	5 GHz	99.4	97.4	100.0
0.1 – 8 (full)	7.9 GHz	99.3	97.4	100.0

All 16 S-parameters. Windows centred on the winning 2-4 GHz region.

1-5 GHz (4 GHz bandwidth, 3 GHz centre) recovers full-band accuracy on every phantom, including 100% on F5. Going wider adds nothing; going narrower costs F5 3-7 points. Half the bandwidth, zero accuracy cost.

Placement and width, at both hardware extremes



Solid = all 16 S-parameters · dashed = single antenna (S11 only). Left: where the 2 GHz window sits. Right: accuracy vs bandwidth around 2-4 GHz.

STEP 3 · COMBINED REDUCTIONS

The 1-5 GHz band with reduced hardware

Hardware @ 1-5 GHz	Empty	A3 + F4	A3 + F5	F5 @ full band
4 antennas, reflection only	**100.0**	**97.4**	**94.9**	*100.0*
2 antennas (1 & 3), full S	99.3	92.9	92.9	*96.0*
2 antennas, reflection only	99.3	92.9	84.8	*90.9*
1 antenna (S11)	96.1	96.2	69.7	*75.8*

The reductions compound on F5: a band cut that is free with the full array costs 3-6 points once antennas or transmission paths are also removed. Best reduced operating point: 4 antennas, reflection-only, 1-5 GHz = 95% on the hardest phantom.

STEP 4 · ROBUSTNESS CHECK

The same sweep with one antenna confirms the band

Band (GHz)	Width	Empty	A3 + F4	A3 + F5
0.1 – 2	2 GHz	87.6	43.6	25.3
2 – 4	2 GHz	**97.4**	**92.9**	**62.6**
4 – 6	2 GHz	95.4	79.5	52.5
6 – 8	2 GHz	98.7	83.3	53.5
0.5 – 5.5	5 GHz	98.0	95.5	74.8
0.1 – 8 (full)	7.9 GHz	98.0	96.8	75.8

S11 only. Note 0.1-2 GHz: fine with the full array (90.9 on F5), catastrophic alone (25.3).

2-4 GHz wins again, so the band choice is robust; but with one antenna there is no knee - accuracy climbs to ~5 GHz. Antennas and bandwidth are partially interchangeable information budgets.

RECOMMENDATION

Chip band specification

Spec: 1-5 GHz

4 GHz bandwidth, 3 GHz centre. Full-band accuracy on every phantom with the array; the anchor for any hardware configuration.

Fallback: 1.5-4.5 GHz

3 GHz bandwidth costs only 1-3 points on the hardest phantom; 2-4 GHz is the floor (-7 on F5).

Avoid: above 5 GHz and below 1 GHz alone

High bands carry no F5 information (insert attenuation). Pure low band collapses without array diversity.

Design principle

Antennas and bandwidth trade against each other. Cut one aggressively, not both: 4-ant reflection-only @ 1-5 GHz keeps 95% on F5.

All results: CNN classification, raw S-parameters, leave-one-session-out, measured A3 phantom data. Details in results/ and docs/CHANGELOG.md.

PART 2 · TRYING TO BREAK IT

New constraint, new question: where does it fail?

Follow-up task: stay entirely below 4 GHz (easier chip) and push the bandwidth down until the model breaks. That needs a definition of 'broken':

DEGRADED	mean below 90% of that phantom's full-band accuracy
UNSTABLE	fold-to-fold sigma of 10 points or more: works one session, fails the next
BROKEN	mean below 50%: wrong more often than right (the headline break)
DEEP	mean below 25%: approaching uselessness (still ~10x chance)

Method: descend bandwidth 3 -> 2 -> 1 -> 0.5 -> 0.25 -> 0.1 -> 0.05 GHz with the best placement at every width, plus a nine-window placement scan at 0.25 GHz. All below 4 GHz.

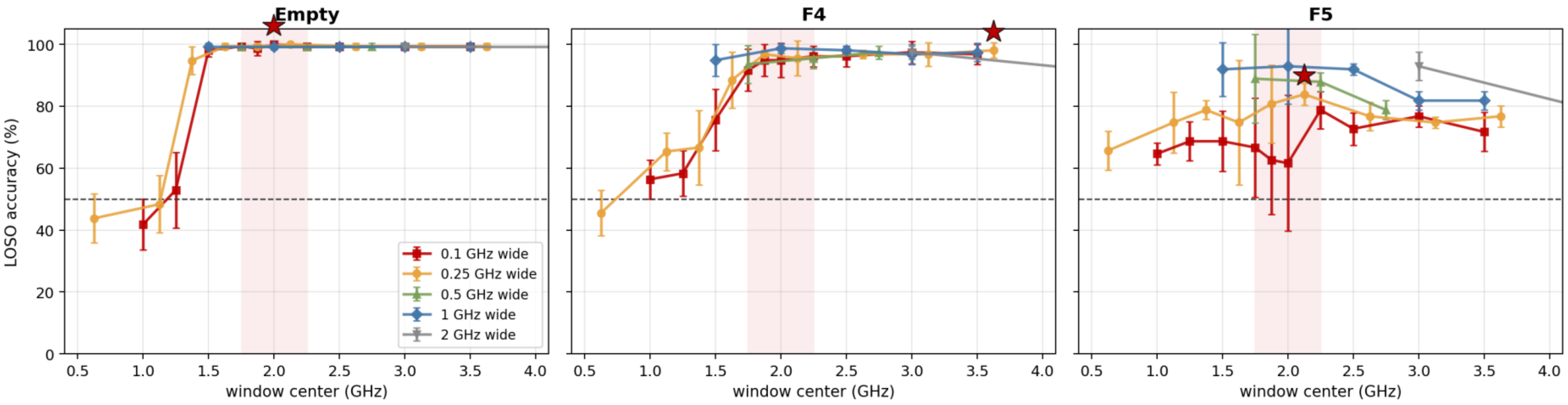
A 0.25 GHz window: placement is everything

Window (GHz)	Empty	A3 + F4	A3 + F5
0.5 - 0.75	43.8 ± 7.9	45.5 ± 7.4	65.7 ± 6.3
1 - 1.25	48.4 ± 9.3	65.4 ± 6.1	74.7 ± 9.7
1.25 - 1.5	94.8 ± 4.5	66.7 ± 12.0	78.8 ± 3.0
1.5 - 1.75	99.3 ± 1.1	88.5 ± 9.0	74.7 ± 20.2
1.75 - 2	99.3 ± 1.1	96.8 ± 1.3	80.8 ± 12.6
2 - 2.25	**100.0 ± 0.0**	**95.5 ± 5.7**	**83.8 ± 3.5**
2.5 - 2.75	99.3 ± 1.1	96.8 ± 1.3	76.8 ± 4.6
3 - 3.25	99.3 ± 1.1	96.8 ± 3.8	74.7 ± 1.7
3.5 - 3.75	99.3 ± 1.1	98.1 ± 2.5	76.8 ± 3.5

Best slot: 2-2.25 GHz (highlighted; also far more stable than 1.75-2). Red rows: below ~1.25 GHz even the EMPTY phantom breaks the 50% line. Bandwidth is survivable; bad placement is not.

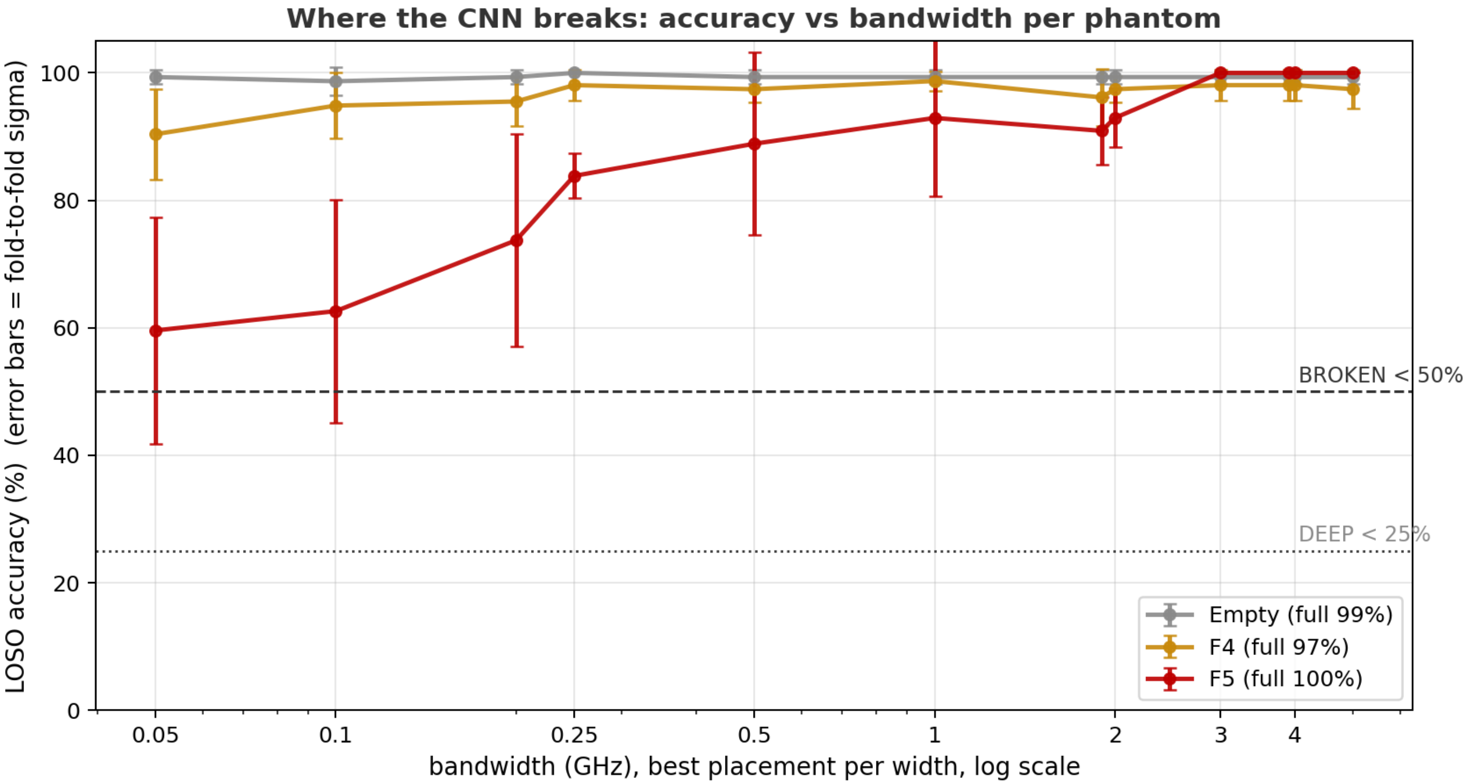
Each phantom has its own best frequency

Where the information lives: train on ONLY one window, slide it across the spectrum. Star = best narrow slot per phantom (each is different)



Sufficiency scan: train on ONLY one window and slide it across the spectrum. Best narrow slot differs by phantom: empty ~2.0 GHz (indifferent above 1.45), F4 ~3.0 (small target rewards higher frequency), F5 ~2.2-2.3 (large lossy insert forces low). The descent's 1.85-1.9 GHz slot was NOT F5's best (62.6 there vs 78.8 at 2.2-2.3), so true narrow-band floors are even higher. Pick the chip band for the worst-case target.

Best placement at every width, down to 0.05 GHz



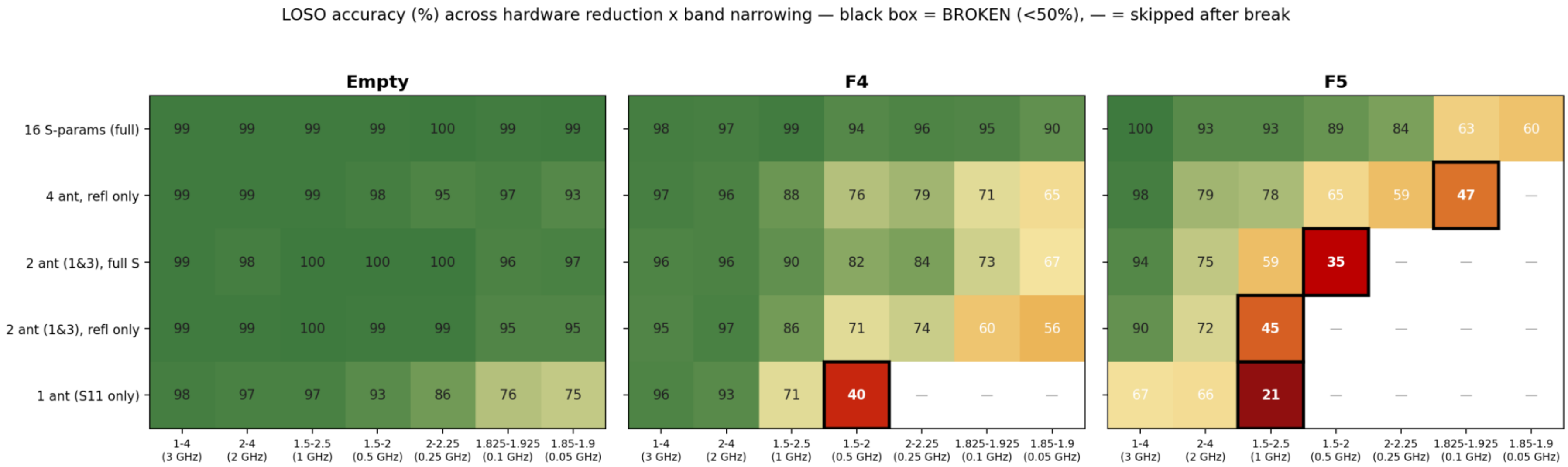
PART 2 · VERDICT

Where each phantom breaks

Failure tier	Empty	A3 + F4	A3 + F5
Degraded (<90% of full band)	never	never	**0.5 GHz**
Unstable (fold sigma >= 10)	never	never	**1 GHz**
Broken (<50%)	never	never	**never**
Floor at 0.05 GHz (50 MHz)	99.3 ± 1.1	90.4 ± 7.1	**59.6 ± 17.8**

With all 16 S-parameters and a well-placed window, bandwidth alone cannot push any phantom below 50%: even a 50 MHz sliver holds 99 / 90 / 60. F5 fails softly (degraded, then session-unstable below 1 GHz). Breaking for real requires combining the band cut with hardware reduction; that adaptive descent is Part 3.

The full break map: 5 hardware levels x 7 bands



Adaptive descent: bands narrow left to right at the best placement per width; once a phantom scores below 50% (black box) it stops descending at that hardware level. Blank cells were skipped after the break. Empty stays green everywhere.

PART 3 · VERDICT

Where each phantom finally breaks (<50%)

Hardware	Empty	A3 + F4	A3 + F5
16 S-params (full array)	never	never	never (floor 60)
4 antennas, reflection only	never	never	**0.1 GHz (47%)**
2 antennas (1 & 3), full S	never	never	**0.5 GHz (35%)**
2 antennas (1 & 3), reflection only	never	never	**1 GHz (45%)**
1 antenna (S11 only)	never (floor 75)	**0.5 GHz (40%)**	**1 GHz (21%)**

- Difficulty ladder is perfectly ordered: empty never breaks (75% even on S11 + 50 MHz); F4 breaks only at the most extreme cut; F5 breaks at every reduced-hardware level.
- Antennas and bandwidth trade against each other: each hardware cut moves F5's breaking point to a wider band (0.1 -> 0.5 -> 1 GHz).
- Chip takeaway: with 2 antennas keep at least ~2 GHz of band (in the 1-4 GHz region); with the full array, bandwidth is nearly free.